# Medical Clinic Digital Twin (BLE-Based Patient Tracking)

## Milestone 1


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**Central Link**

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# I. Problem Statement and Motivation

In busy medical environments, managing patient flow and reducing wait times is a significant operational challenge. Traditional methods of tracking rely on manual check-ins, which often fail to capture real-time bottlenecks or the precise movement of patients between departments. This lack of visibility can lead to overcrowded waiting rooms, inefficient use of clinical staff, and a poor overall patient experience.

The **Medical Clinic Digital Twin** addresses this by automating the data collection process. By understanding exactly how long a patient spends in each area—from the reception to the examination room—hospital administrators can make data-driven decisions to optimize workflows and improve service delivery.

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## Use Case: Real-Time Patient Tracking

The system is designed for a clinical setting where patients are issued a low-power Bluetooth Low Energy (BLE) tag upon arrival. These tags broadcast a unique identifier as the patient moves through the facility.

- **Who are the users?**
    - **Clinic Administrators:** They use the data to identify bottlenecks and optimize staffing.
    - **Medical Staff:** They gain awareness of patient locations to streamline transitions between consultations.
    - **Patients:** They benefit indirectly from reduced wait times and a more organized environment.

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## Expected Outcomes

The implementation of this IoT system is expected to deliver the following results:

- **Real-Time Visualization:** A digital dashboard that mirrors the clinic's physical layout, showing the current location of every active tag.
- **Workflow Analytics:** Automatic calculation of average wait times and "stay durations" in specific clinic zones.

- **Anomaly Detection:** Use of machine learning to identify unusual movement patterns or unexpected delays that require immediate attention.
- **Operational Efficiency:** Data-backed insights that allow the clinic to handle higher patient volumes without increasing wait times.

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## II. System Architecture

The system follows a tiered IoT architecture designed to capture physical movement and translate it into a synchronized digital twin. The architecture consists of the following layers:

### System Components and Communication

- **End Devices (Perception Layer):** Low-power BLE tags act as the data source. These devices operate in "advertising-only" mode to broadcast a unique ID and a timestamp, ensuring high energy efficiency for mobile tracking.
- **Gateways/Listeners (Edge Layer):** Stationary ESP32 devices are placed at clinic doorways. These listeners act as BLE Observers, scanning for advertising packets, measuring the Received Signal Strength Indicator (RSSI), and appending a gateway ID to the data.
- **Cloud Infrastructure (Network & Processing Layer):** Data is transmitted from the gateways to a central cloud server via WiFi/Ethernet. The cloud hosts an API that collects data from all gates, performs time synchronization, and manages a database to resolve location conflicts based on the strongest RSSI and most recent timestamp.
- **User Interface (Application Layer):** A web-based dashboard fetches processed data from the cloud. It visualizes the clinic layout and utilizes machine learning to classify movement patterns and identify bottlenecks in real-time.

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### Architecture Highlights

- **Communication Technologies:** The system utilizes **BLE (Bluetooth Low Energy)** for the short-range link between patients and gates due to its low power

consumption, and **WiFi/ MQTTS** for the long-range link between gates and the cloud.

- **Data Integrity:** To ensure all gateways are synchronized, the cloud periodically pushes time updates to the edge devices, ensuring the sequence of patient movement is recorded accurately.
- **Security:** A security layer is implemented between the gateways and the cloud API to ensure that patient tracking data remains confidential and protected from unauthorized access.

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## III. Hardware Components

To implement the medical clinic digital twin, the following hardware devices are required. Each component plays a specific role in capturing and transmitting patient movement data.

| Device Name | Purpose in the System | Communication Interface |
|---|---|---|
| **BLE Advertising Tags** | Worn by patients to broadcast a unique ID, acting as the primary data source. | BLE Advertising (2.4 GHz) |
| **ESP32 Microcontrollers** | Installed at doorways to scan for BLE signals, process RSSI values, and act as gateways. | BLE and WiFi/Ethernet |
| **Central Cloud Server** | Receives data from all ESP32 gateways, manages the database, and hosts the visualization API. | MQTTS / HTTPS / REST API |

| Device Name | Purpose in the System | Communication Interface |
|---|---|---|
| **Display Dashboard** | A web-based interface (accessible via PC or tablet) to visualize the clinic layout and patient flow. | HTTP/WebSockets |

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## IV. Communication Protocols

The system utilizes a combination of short-range and long-range protocols to balance power efficiency with reliable data delivery.

- **BLE (Bluetooth Low Energy):** Selected for the link between the patient tags and the gateways. BLE's advertising mode allows the end devices to operate with extremely low power consumption, which is essential for battery-operated wearable tags.
- **MQTT over TLS (MQTTS):** Between the gateways (listeners) and the cloud server. MQTT is a lightweight publish-subscribe protocol ideal for real-time IoT telemetry. To meet clinical privacy standards, **TLS (Transport Layer Security)** is implemented to encrypt the data stream, ensuring that patient IDs and location data are protected from interception during transit over the clinic's network.
- **NTP (Network Time Protocol):** For gateway time synchronization. To ensure that the "earliest timestamp" logic works correctly when reconciling data from multiple gates, all gateways must have a synchronized clock. NTP allows each gateway to maintain a precise sub-second accuracy relative to the cloud server.
- **JSON (JavaScript Object Notation):** Data format for the MQTT payload. JSON provides a structured, human-readable format that is easily parsed by both the edge devices (ESP32) and the cloud-based web application.

- **HTTPS:** Employed as the primary transport protocol for the API to ensure that all data transmitted between the clinic hardware and the cloud server is encrypted and secure.

| Link | Protocol | Security/Feature |
|---|---|---|
| **End Device → Gateway** | BLE Advertising | Low Power / Connectionless |
| **Gateway → Cloud** | MQTT | Lightweight / Real-time |
| **Data Encryption** | TLS 1.2/1.3 | End-to-End Encryption |
| **Cloud → Dashboard** | WebSockets / HTTPS | Live Updates |

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## V. Expected System Workflow

The operation of the Medical Clinic Digital Twin follows a structured sequence to ensure real-time accuracy and data integrity. The step-by-step process is as follows:

1. **Tag Broadcast:** Wearable BLE end devices (tags) continuously broadcast advertising packets containing a unique ID.
2. **Edge Detection:** As a patient moves, gateways (listeners) at doorways capture these advertising messages.
3. **Local Processing:** The gateways measure the RSSI (signal strength) and attach a local timestamp and gateway ID to the packet.

4. **Cloud Transmission:** The gateways forward the data to the central cloud server via a secure MQTTS API.
5. **Data Reconciliation:** The cloud API analyzes incoming data from multiple gates. If a tag is detected by two gates simultaneously, the system selects the entry with the strongest RSSI and the earliest timestamp to determine the precise location.
6. **Time Synchronization:** To prevent data drift, the cloud periodically sends time-sync packets back to the gateways.
7. **Analysis & Visualization:** The cloud database updates the digital twin layout, and machine learning models classify the movement patterns.
8. **User Output:** The web application displays the real-time location and workflow analytics (wait times and bottlenecks) to clinic administrators.